

Notice of Intent to Review Marketplace Solutions

SHOPPING NOTICE: DARPA-SN-26-69

MARKETPLACE: Defense Advanced Research Projects Agency (DARPA) Expedited Research Innovation System (ERIS)

REQUIRING ACTIVITY: DARPA, Microsystems Technology Office (MTO), Next-Generation Microelectronics Manufacturing (NGMM) Program

NOTICE DATES: April 16, 2026 – July 31, 2026

TOPIC AREA: Strategic Systems

1.0 PURPOSE

Between the dates listed above, the DARPA NGMM program¹ will be actively reviewing the ERIS Marketplace for innovative, post-competitive, awardable solutions that exercise the early-stage version of the NGMM 3D Assembly Design Kit (3D-ADK) for mixed-material 3D-heterogeneously integrated (3DHI) microsystem design with commercial Electronic Design Automation (EDA) tools. This is an active and immediate search for solutions. Submissions will be reviewed on a rolling basis, and the Government reserves the right to make one or more awards prior to the closing date of this notice.

2.0 PROBLEM & SCOPE

The NGMM program is developing new EDA capabilities, and the first comprehensive multi-physics enabled 3D-ADK, for the design of advanced mixed-material 3DHI microsystems. Enabling efficient EDA workflows, built around a robust 3D-ADK, is essential to the NGMM program and the sustainability of the resultant high-mix/low-volume pure-play advanced packaging capability. DARPA is interested in awardable solutions to exercise and evaluate the early-stage release of the 3D-ADK and EDA workflow.

- **Current Obstacle:** For 3DHI designs, existing EDA tools and ADK infrastructure are insufficient to achieve architectural exploration studies where system level trades and performance can be quickly simulated with relative precision and detailed design capabilities where co-design activities with concurrent analysis and optimization can be employed for full assembly sign-off.
- **Desired Research Focus:** We are seeking solutions, informed by a relevant 3DHI design aligned with the NGMM technology roadmap^{2,3} and that will comprehensively exercise the NGMM 3D-ADK and EDA capabilities, to support architectural trade studies and verify design functionality and performance to the sign-off phase. Figure 1 shows the planned offerings on the technology roadmap for the NGMM center. As the goal of the effort is to validate the NGMM design infrastructure, proposed solutions should align to the NGMM 3D-ADK as much as is practical.

¹ Holmes, M., et al., “The Defense Advanced Research Agency’s (DARPA) Next Generation Microelectronics Manufacturing (NGMM) Program,” International Conference on Compound Semiconductor Manufacturing Technology, New Orleans, May 21, 2025.

² “MTO Spark Tank,” YouTube, <https://www.youtube.com/watch?v=FNpwnqWJd6c>, July 2025.

³ “NGMM Summit,” Defense Advanced Research Projects Agency, <https://www.darpa.mil/events/rewinds/ngmm-summit/presentations>, October 2025.

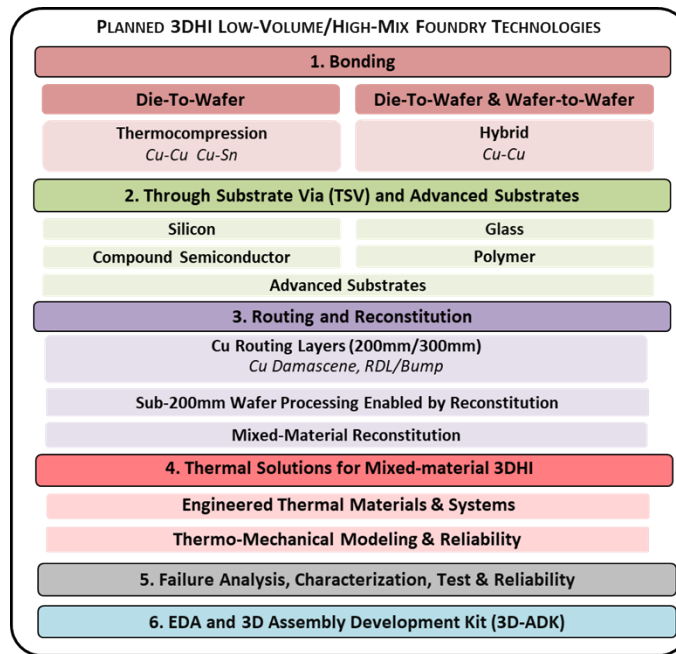


Figure 1: NGMM offerings to enable design and manufacturing of 3DHI microsystems. Supported materials include Si, glass, GaN, InP, GaAs, SiC, and yttrium iron garnet.

3.0 DESIRED CAPABILITIES AND EXPERTISE

While *all submissions will be evaluated against the standard criteria published in the ERIS Master Solicitation*, the DARPA NGMM program is specifically looking for solutions that demonstrate strength in the following areas. These are not new evaluation criteria but rather highlight the specific attributes of interest for this opportunity.

- **Advancing the State of the Art (Technical Innovation):**

Proposed solutions should reflect a next-generation, innovative 3DHI design that provides substantial performance benefits over conventional 2D designs composed of a single material. The 3DHI design is expected to have two or three layers of components vertically integrated using the planned NGMM center offerings shown in Figure 1. The most compelling solutions will be those that align with the NGMM integration capabilities, 3D-ADK, offering innovative methods to enable the comprehensive evaluation and validation of design flows and models.

- **Team Capability (Expertise and Vision):**

We are seeking teams with significant expertise in mixed-signal, mixed-materials design with modern, multi-physics EDA tools. Teams should have a proven track record of designing microelectronics utilizing compound semiconductors and silicon, as well as working with multiple foundries and multiple design kits to develop advanced microsystems. Additionally, experience with 3DHI or mixed-materials 2.5D integration is strongly preferred.

- **Defining the Problem & Current State of the Art:**

Strong submissions will demonstrate a deep understanding of the problem space, in which existing EDA and ADK infrastructure have limited support for comprehensive 3DHI device simulation and verification on a reasonable timescale. Proposers should clearly define the scope of the technical challenge their 3DHI design addresses and compare their proposed solution to the current state of the art.

- **Impact (Defense and Commercial Use Case):**

Submissions should discuss how the proposed solution will demonstrate the significant performance advantages of 3DHI microsystems and improve the robustness and effectiveness of the NGMM 3D-ADK and EDA workflows through a candidate design flow aligned with directly relevant Department of War and/or dual use applications.

4.0 HOW TO RESPOND

This notice is a call for solutions to be flagged for review.

For Existing Awardable Solutions:

If you have a solution already in the ERIS Marketplace that aligns with this need, send an email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-69 - [Your Company Name]
- Body: The Title and Video URL of your existing awardable solution.

For New Submissions:

If you have a new solution, please submit it to the 'Strategic Systems' Topic Area via the ERIS portal. The full description for that topic area is:

Developing advanced technologies for improved resilience, efficiency, and effectiveness of strategic systems. This topic area supports technologies for critical infrastructure and military command-and-control systems (networks and applications across strategic, command, operational, and tactical edges).

To ensure it is associated with this notice, please use the following format for your video title:

- Video Title Format: DARPA-SN-26-69: [Your Solution Title]
(Example: DARPA-SN-26-69: AI Accelerator Analysis Tool)

After submitting, please also send a notification email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-69 - [Your Company Name]
- Body: The Title of your new submission and its Date of Submission.

Solutions submitted during this window that are deemed "awardable" through the standard ERIS assessment process will be included for consideration by NGMM.

5.0 RESOURCES

- ERIS Master Solicitation: [SAM.gov Link](#)
- To Submit a New Solution: [DARPAConnect ERIS Portal](#)